

VIKASIT NAGPUR HACKATHON 2026

PROBLEM STATMENT

uneven distribution of traffic over planning authorities jurisdiction

Expected solution

simulation to create a traffic management during traffic hours (9am to 12 noon) and 4pm to 7pm

SOLUTION

SMARTFLOW

TRACK

Web Dev

TEAM NAME

Coders 2.0

TEAM MEMBERS

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Problem Statement and Solution

Detailed Solution

- **AI powered traffic analysis**
- **Real time traffic monitoring**
- Interactive traffic map visualization
- **Simulation based traffic management**
- **Automated report generation**

Problem Statement addressed

- **Identifies uneven traffic distributuion**
- Optimizes traffic flow
- Supports faster decision making
- Reduce traffiuc congestion
- Detects congested roads during traffic hours



Innovation and uniqueness

- AI based traffic recommendations
- **Traffic simulation before implementation**
- Before vs after traffic comparison
- Dynamic strategy selection
- **Data driven smart city solution**

By combining real time analysis ,AI recommendations,and traffic simulation ,the systemhelps authorities optimize traffic flow & support data driven decision making

- uneven traffic distribution
- traffic analysis
- simulation
- management strategy
- optimized traffic
- measurable improvement

TECHNICAL APPROACH

Technology Stack

Frontend



Backend



Framework



Database



Deployment



External APIs



Implementation Plan

1

login and dashboard

- Secure access for authorities
- Display real time traffic data

2

select peak hour and identify congested roads through map

Choose the required time slot
Detect traffic hotspots using the interactive map

3

Open traffic simulation and select traffic management strategy

- Create a virtual traffic environment
- Select an appropriate traffic control strategy

4

Run simulation and view traffic flow

- Execute the selected strategy
- Visualize traffic movement and density

5

compare before vs after and view AI recommendations

- Evaluate the impact of strategy
- Receive the AI based optimization suggestions

6

Apply/save strategy and generate report

- Implement the best performing strategy
- Save results and generate a detailed report



IMPACT AND BENEFITS

Impact



- Reduced traffic congestion
- Improved road safety
- Faster emergency response
- Data driven decision making
- Environmental sustainability
- Reduce delays during emergencies

Impact

Benefits



- Better management for public infrastructure
- Real time monitoring of traffic conditions
- Improved public trust in government services
- Support for sustainable urban development and smart city

Benefits

Smart
city

Smarter
governance

Smart
mobility



RESEARCH AND REFERENCES

- google map stored data for analysis
- intelligent urban traffic management system(elsevier 2022)
- real time traffic managment in smart cities (MDIP,2026)
- Artificial intelligence in urban transportation system
- traffic prediction using artificial intelligence (elsevier 2022)
- research papers